



Non-invasive anaemia detection by examining palm pallor: A smartphone-based approach

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ABSTRACT

Anaemia, caused due to lack of blood haemoglobin level, is one of the most common diseases that affects billions of people across the world. According to WHO statistics, India is one of the developing countries with highest prevalence of anaemia. Conventional invasive methods are cost-prohibitive and difficult to administer globally which essentially demands non-invasive, inexpensive and reliable solution. This work targets to combine state-of-the-art computational approaches with the idea of age-old practice of rough estimation of blood haemoglobin level by observing pallor in palm to develop a non-invasive anaemia detection system. The proposed system works with the basis of inducing changes in palm pallor with suitable pressure application and release, measuring the rate of change of colour and performing time-domain analysis thereof to correlate with blood haemoglobin concentration. The video of colour changes in palm induced through a customised device, is captured using smartphone camera and is processed and analysed through a set of image processing and analysis techniques. Estimated parameters are provided as input to the ensemble-based prediction models comprising of regression and multi-layer perceptron. The proposed system ensures a sensitivity of 93% with MSE and RMSE of 0.701 and 0.698, respectively, determined on palm pallor video samples of 131 participants.

1. Introduction

Anaemia is a condition in which the number of red blood cells decreases and the oxygen-carrying capacity of blood becomes insufficient to meet the physiological needs [1]. This condition may also vary with altitude, smoking, age, sex, and other health status, like pregnancy. This condition leads to anaemia disease or deficiency of haemoglobin levels in the human body. The primary causes of anaemia are thought to be due to iron deficiency only. However, several other reasons, such as foliate, chronic inflammation, vitamin B12 and vitamin A deficiencies, parasitic infections, and inherited disorders, etc., may also cause anaemia. In its severe form, it causes weakness, fatigue, drowsiness, and dizziness, etc. Children and pregnant women are particularly vulnerable to anaemia. According to National Family Health Survey (NFHS) statistics, 67.1% of children and 57% of women suffered from anaemia and the percentage is higher at rural areas. These were among the findings of NFHS-5 conducted from June 2019 to April 2021, carried out by the Union Ministry of Health and Family Welfare of India.

Anaemia may remain undetected in many individuals due to minor symptoms and especially in underdeveloped areas due to a lack of medical facilities, health care, and self-awareness. The traditional

approach to detecting whether a person is anaemic or not considers the evaluation of haemoglobin concentration in blood. In this process, blood sample is collected from an individual and then the haemoglobin concentration is determined using a hematocrit test. This test is used to evaluate the percentage volume of red blood cells present in the blood. The whole invasive process requires laboratory setup and laboratory testing, which is time-consuming and may expose the health workers to blood-borne infections. Nowadays, smartphone or smartwatch-based medical devices are becoming popular to estimate health-related parameters in non-invasive ways [2–4]. For example, heart rate monitoring, sleep monitoring, pulmonology, fall detection, blood oxygen (SpO_2) monitoring, ECG monitoring, etc., are among such devices. Thus, there is a requirement for efficient non-invasive techniques for the detection of anaemia thereby, to avoid the complication induced in the invasive methods.

Several age-old practices may include checking of eye and nail pallor for a rough estimation of a person's anaemic condition, however, most scientific procedures involve taking a patient's blood sample and then determining the haemoglobin count. This haemoglobin count gives a direct estimation of the person's health condition. However,

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the invasive procedures come with their own set of challenges. It is not possible to expect every individual to opt for a haemoglobin count test in rural India due to cost factors and other administrative issues. The additional taboo attached to invasive procedures makes the task even more difficult. Keeping in mind the severity of the disease and the associated difficulties concerning the majority of the population, it becomes a task of paramount importance to design a robust and fast procedure for smart non-invasive anaemia detection system. Thus, the proposed work suggests a procedure that combines the age-old non-invasive techniques (for a rough estimation of blood haemoglobin level) with cutting-edge Artificial Intelligence (AI) techniques for accurate estimation of blood haemoglobin level, thereby alleviating the problems of invasive procedures.

The objective of this paper is to design and develop a non-invasive palm pallor-based anaemia detection system that uses a smartphone-based application. The contributions of the proposed work are summarised as follows:

- Design and development of a portable, reliable, user-friendly, non-invasive system that ensures prediction of closely approximated haemoglobin levels for patient care at a minimal cost
- Use of dynamic physiological palm pallor evidence for analysing and detecting anaemia through estimation of the haemoglobin level of an individual
- Application of image processing techniques to carry out the tasks, such as image registration, region of interest (ROI) extraction and image quality assessment to improve the performance of anaemia detection system
- Uses of some novel ensemble-based prediction frameworks comprising of machine learning models, such as regressions and neural networks, for accurate estimation of the blood haemoglobin levels
- Accurate evaluation of whether a person is anaemic or not, along with a fine grain classification of anaemia according to WHO standards
- Use of a range of performance metrics, viz. confusion matrix, risk analysis graph, MSE, RMSE, Limits of Agreement (LOA) to better understand the performance of the anaemia detection system while determining the haemoglobin levels.

The rest of the manuscript is organised as follows. Section 2 presents a comprehensive study on the existing works that use different physiological evidences to estimate haemoglobin levels. The proposed system has been detailed out in Section 3 followed by evaluation of the proposed system made in Section 4. The last section concludes the proposed work and briefly discusses the possible future works.

2. Literature survey

This section provides a comprehensive study on the existing works on anaemia detection methodologies. As reported in the literature, there exist some invasive and non-invasive methods for anaemia detection. Invasive methods use the blood samples collected from individuals and is then tested in the laboratory for estimating haemoglobin measurements for anaemia detection. HemoCue [5] includes a haemoglobin photometer that predicts results more accurately. It is equipped with a battery and is portable, however, expensive. Whereas, WHO colour scale [6], is cheaper as compared to HemoCue [5], and evaluates the haemoglobin levels by comparing blood colour with a colour-matching scale, but with no encouraging end result. This method does not give an accurate estimation. The most commonly used method for evaluating the haemoglobin levels is the gold standard method [7] based on the Cyanmethaemoglobin procedure. This requires a well-equipped laboratory for drying blood with the help of filter paper and redissolving. Also, spectrophotometry is used to measure light absorption, which is a cost-prohibitive method. S. Suner et al. [8] started the evolution of the non-invasive detection method with the help of digital photography

of palpebral conjunctiva. This is not a smartphone-based method. Further, Masimo pulse co-oximetry [9] and Masimo Pronto [10] methods are introduced, based on the spectrophotometric principle, which can accurately estimate the haemoglobin levels. However, these techniques are more expensive.

Non-invasive methods are also used to estimate the haemoglobin levels for anaemia detection and these methods use images of physiological evidences. Santra et al. [11] made a machine vision-based portable, non-invasive, cost-effective, and user-friendly system to measure haemoglobin levels from the redness of the skin of a palm. A sphygmomanometer cuff is used near the wrist, which restricts the blood flow by raising the pressure and then suddenly, the pressure is released. The entire video is recorded for approximately a minute, including 15 s each for pre and post-blood flow restriction. However, the percentage of anaemic and non-anaemic individuals has not been mentioned. Wang et al. [12] designed a mobile application called HemaApp, which allows recording a video under various lighting conditions to monitor the blood haemoglobin concentration with the help of a smartphone's front camera. They perform a chromatic analysis on a fingertip for analysing the colour of blood to estimate the haemoglobin level. Mannino et al. [13] developed a smartphone application for anaemia detection by measuring haemoglobin levels non-invasively. This application allows capturing photographs of nail-beds of each finger of patients. Patients can manually select the region of interest of their nail-bed of each finger. Different features like fingernail, skin colour, and image metadata are extracted and processed using a multi-linear regression with a bi-square weighting algorithm to estimate the haemoglobin level. Karel J Zuzak et al. [14] evaluate the correlation between blood flow in the palm and haemoglobin value in blood of an individual for assessment of sickle cell anaemia. The flow of blood can be evaluated by the colour analysis of a palm and infer the haemoglobin level. They use the visible reflectance hyperspectral imaging technique to quantitate haemoglobin oxygen saturation HbO_2 on skin tissues. V. Bevilacqua et al. [15] proposed a non-invasive technique for haemoglobin detection based on a specific conjunctiva region image. They have designed a wearable device that is easy to use by patients. This work also focuses on the segmentation of the specific conjunctiva region, which mostly correlates with the hematic concentration of Hb . The estimation is based on a^* axis (runs from left to right) of LAB colour space, finding an average correlation with haemoglobin values in the blood. Prakhar Jain et al. [16] discussed a novel machine learning model using a neural network for the detection of anaemic patients from the use of eye conjunctiva images. Md. Moin Uddin Atique et al. [17] measures haemoglobin through the inner lower eyelid image captured under uniform lighting conditions. They consider the manual region with 10×10 pixels and avoid the area which gives more reflection. The average red value pixel for the selected region is evaluated. Also, an eyelid conjunctiva data set is available for estimation of haemoglobin [18]. Some pre-processing and segmentation process is applied on this dataset by Giovanni Dimauro et al. [19] and Sivachandar Kasiviswanathan et al. [20]. Giovanni Dimauro et al. [19] uses graph partitioning to maximise the correlation between colour properties and summarises the segmentation process into three phases: dimension reduction by K-means clustering approach, grouping with region adjacency graph and a final ROI extraction. Sivachandar Kasiviswanathan et al. [20] applied image augmentation along with angular rotation, horizontal flipping, Gaussian blur and noise addition as pre-processing steps. The U-Net based conjunctiva segmentation model is applied for segmentation.

There exist a few related works in anaemia detection based on palm pallor or palm crease pallor. However, none of these works, as reported in the literature, apply any image processing techniques on the image or video data captured for analysis, instead, they examine the pallor manually by clinical signs. G. Santra [21] evaluates the value of palmar creases for the detection of severe anaemia. As in severe anaemia cases, palmar creases become as pale as the surrounding skin. D. Regina

et al. [22] measures the correlation of pallor with haemoglobin levels. The clinical data profile of anaemia is created with school children of rural Telangana consisting of conjunctival pallor, skin pallor, and palm pallor. S.P. Luby et al. [23] use clinical signs to diagnose anaemia in children of Africa. They collected blood samples and captured the pallor in the tongue, eye conjunctiva, nail bed and palm from 1104 children. R. S. Manesh et al. [24] studied the clinical images of palm crease pallor of hands. The pallor of the palmar creases in the examiner's hand is compared to the subject's hand to make rough estimation of haemoglobin level.

The invasive schemes as reported in the literature are highly cost-intensive, time consuming and are difficult to apply globally due to resource constraints and other administrative issues. Whereas, most of the non-invasive schemes use digital images as data with colour channels as common features for estimation of haemoglobin level. However, to overcome the limitations of accuracy issues of static samples (image), the proposed methodology considers dynamic samples (video) to better capture the rate of change of colour with the target to enhance accuracy of the system as compared to other existing solutions.

3. Overview of proposed non-invasive anaemia detection system

This section describes the overall system of the proposed non-invasive anaemia detection technique based on palm pallor, which consists of the following sub-systems:-

3.1. Data acquisition

A hardware prototype model has been developed indigenously to acquire the palm evidence from palm video photography and further, use of the evidence for measuring the haemoglobin levels. It uses uniform pressure on the entire palm with the blood pressure arm cuff to decolourise the palm by obliterating the entire venous drainage and a major part of arterial supply to deoxygenate the stagnant blood there. Thereafter, the blood flow is allowed suddenly on the palm to get oxygenated and stabilised by spreading oxygen to suffocated palm tissues. The timestamps of various stages of palm compression and release can be directly obtained from the video photography of the palm. The inner side of the palm instrument is painted with black colour, which helps to block the ambient light by avoiding the reflection of light. Further, the influence of ambient light is managed by providing constant illumination inside the videography box with the help of LED light strips. The LED light strips are placed over the inner top of the instrument to provide constant illumination during the acquisition. Fig. 1 shows the device prototype along with the placement of the hand. A smartphone camera is used to capture the video of the palm under a specific environmental setup by fixing the palm firmly and comfortably. The smartphone camera starts recording the video when palm pallor is under no pressure. After 10 s, pressure application starts and reaches 20 mm Hg more than the systolic pressure of an individual and thereafter, it retains the constant pressure for 40 s. Then the pressure is released after a certain period of time to observe the stability of blood distribution in the palm. The entire event is video photographed until the palm pallor reaches its normal colour. The total time duration of the video is of 1 min and 40 s. The videography considers only bare hand (palm) with no artificial colour coating. With the hardware model, video photography of palm pallor from 131 participants have been acquired at two different places in Durgapur, India. 131 individuals have contributed their blood samples for clinical measurement. Among 131 participants, 17 individuals have contributed at the Parulia Health Center, Matrisadan, Durgapur, and remaining 114 individuals have contributed at the ESI Hospital, Durgapur. Redmi K20 Pro Smartphone, equipped with 48 Mega-pixel Sony IMX586 sensor with $f/1.75$ aperture and 1.6-micron pixels, is used for capturing the video. All the videos with resolutions of 1920×1080 pixels with a frame rate of 30 frames per second (fps), are recorded. During data



Fig. 1. Device prototype.

collection, the pressure is applied below the left-hand elbow of each participant with the help of a blood pressure measuring pressure cuff. To fix the palm on the base situated beneath the camera focus with a permissible distance, velcro strips are used. To prevent the movement of the subject during data collection, the subject is requested to sit comfortably and steadily as well as not to speak during the experimentation. Table 1 shows the specifications of the collected data. The prior approval from the Institutional Ethical Committee has already been obtained before commencement of data collection process. Also, written consent from all the participants have been obtained prior to the study.

3.2. Pre-processing techniques

The image preprocessing sub-system consists of a sequence of crucial steps, viz. frame extraction from video data, image registration for alignment, region of interest (ROI) extraction and image quality assessment for analysis of video data collected from the participants.

3.2.1. Image registration

The image registration process is used to map the different instances of the same image when the image is acquired from different viewpoints, different time stamps and different sensors. The present work uses the video sample of a palm pallor to extract a number of frames and these frames are considered to be instances of the same palm pallor image where the changes in blood distribution are observed at different Spatio-temporal instances. To align these instances with the reference frame in the same coordinate system facilitates the process of image registration and further accumulates distinctive and rich information about the colour changes on palm pallor, thus helping to estimate the haemoglobin levels of individuals accurately.

The proposed system exploits the image registration process to align the extracted frames with the first frame of palm pallor considered to be the reference frame. The features of the reference frame and any input frame among the rest of the frames are considered for alignment in the coordinate system, and the (x, y) coordinates of the matched keypoints are stored. The matched keypoints between two image frames (reference and input frames) are determined with the help of the ORB (Oriented FAST and Rotated BRIEF) algorithm [25]. Then the motion model, which consists of shearing, scaling, rotation, and translation, has been applied for estimating the warp matrix, which corresponds to composite transformation. Finally, the optimised warp matrix is determined by minimising the variations between two frames with the help of the Enhanced Correlation Coefficient Maximization algorithm [26], and the warp matrix is then applied to the input image to obtain the registered image. The alignment is also required on a fixed palm due to variations in pressure i.e., while applying a pressure

Table 1

Dataset description

Participant information			Age (in years)		Gender			Blood Hb (in g/dL)
Total	Anaemic	Non-anaemic	Range	Average	Male	Female (Non -pregnant)	Female (Pregnant)	Range
131	100	31	18-79	38.09	45	63	23	7.8–15.4

**Fig. 2.** Palm image registration.

and releasing the pressure, sometimes the palm may be misaligned slightly. Also, while applying pressure on aged subject it might result into tremor. So, to correct the effect of misalignment while analysing the image, the alignment is performed by applying image registration algorithm. Algorithm 1 shows the process of image registration for the proposed system. Fig. 2 shows the output after applying Algorithm 1.

Algorithm 1 Palm Image Registration

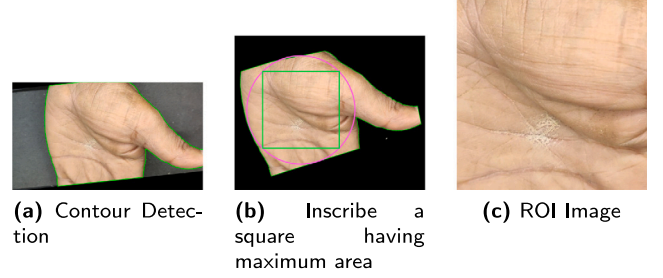
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Input : Video Data
Output : Registered Frames
1:  $reg\_frames = []$ 
2: For each video  $frames = [f_1, f_2, \dots, f_N]$ 
3:  $im1 = im[0]$  // Frame extraction
4: for  $<i = 1 : N>$  do
5:  $im2 = im[i]$  // Read first frame  $f_1$  as a reference frame
6: Obtain the grey level image for  $im1$  and  $im2$ .
7: Apply feature descriptors and motion model between  $im1$  and  $im2$ .
8: Allocate space for  $warp\_matrix$  to store the motion model.
9: Estimate the  $warp\_matrix$  by minimising the variation between  $im1$  and  $im2$ .
10:  $registered\_image = warp\_matrix \times im2$ 
11:  $reg\_frames[i] \leftarrow registered\_image$ 
12: end for
13: return  $reg\_frames$ 

```

3.2.2. ROI extraction

As for each frame, it is necessary to keep the focus on the thenar and hypothenar muscle parts of the palm, region of interest (ROI) to be extracted from the palm would carry such significant information and contribute to detecting the anaemia precisely. The ROI detection and extraction consist of a number of steps. Firstly, the lower and upper bounds of skin areas in the YCrCb colour space are defined. Then the area of interest from the image is extracted after applying thresholding, as shown in Fig. 3(a). Afterwards, the algorithm finds the contours from the thresholded image and then it obtains the maximum contour of the area of interest. Then vertical adjustment correction is then performed to make all segmented frames from different subjects to the same orientation, and the Watershed algorithm [27] is applied for estimating the distance to the closest zero pixels, i.e., a boundary for each pixel of the image. The maximum contour area is evaluated with the help of Green's theorem [28], which is used for finding the area. Then draw a circle with a maximum radius ($maxdist$) on the palm image with maximum contour, as shown in Fig. 3(b). At last, inscribe a square with the help of a metric ($half_slide$) that gives the base on a circle that

**Fig. 3.** ROI detection on palm image.

generates the region of interest of a palm, as shown in Fig. 3(c). The metric ($half_slide$) is evaluated as shown in Algorithm 2.

Algorithm 2 Palm ROI Detection

```

Input : Registered Frames
Output : ROI Frames
1: For each video  $reg\_frames = [Rf_1, Rf_2, \dots, Rf_N]$ 
2:  $crop\_frames = []$ 
3: while ( $i \neq NULL$ ) : do
4:  $im[i] = \text{convert RGB to YCbCr image}$ 
5:  $contours = \text{extract skin areas and apply Thresholding}$ 
6:  $largest\_contour = \text{find the maximum contour area by using Green formula}$ 
7: Vertical adjustment correction
8:  $dist\_img = \text{apply Watershed algorithm}$ 
9:  $maxdist = 0$ 
10: for  $<k = 1 : dist\_img.shape[0]>$  do
11: for  $<l = 1 : dist\_img.shape[1]>$  do
12:  $dist = dist\_img[k][l]$ 
13: if  $<maxdist < dist>$  then:
14:  $x = l$ 
15:  $y = k$ 
16:  $maxdist = dist$ 
17: end if
18: end for
19: end for
20: Inscribe circle with the help of  $maxdist$ ,  $x$  and  $y$ .
21:  $half\_slide = maxdist * \cos(\frac{\pi}{4})$ 
22:  $(left, right, top, bottom) = ((x - half\_slide), (x + half\_slide), (y - half\_slide), (y + half\_slide))$ 
23:  $crop\_frames[i] \leftarrow \text{patch of dimension } (top + bottom) \times (left + right) \text{ pixels is selected from image } im$ 
24: end while
25: return  $crop\_frames$ 

```

3.2.3. Image quality assessment

Every extracted frame contains noise, pixel bit value, RGB colour levels, and histogram information, which are useful for analysing video quality. The degradation of image quality affects the image visibility and degrades the colour information available in the image. The present work considers three relevant parameters such as dynamic range, noise, and sharpness of the image for image quality assessment. Dynamic range corresponds to the difference between the maximum and minimum lightness conditions. Noise may be created due to the degradation of image signal caused by external sources and sometimes by sensors,

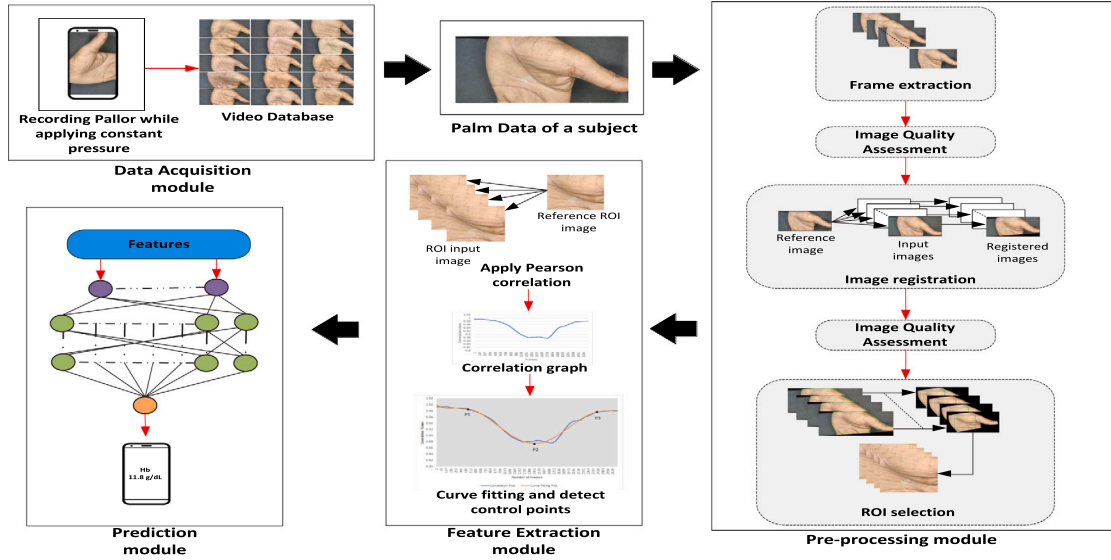


Fig. 4. Overview of the system with feature extraction.

Table 2
ISO standardised metric values.

Metrics	Excellent	Very good	Good
Dynamic range	≥ 2.3	≥ 2.1	≥ 1.9
Noise	< 3	< 4	< 6
Sharpness	≤ 1.05	≤ 1.1	≤ 1.2

and sharpness is a measurement related to the contrast and texture of an image. As per ISO standards [29], a pre-specified range of these metrics validates the image quality. The dynamic range value can be improved by exploiting Bit plane slicing. The pixel intensity value of an image is a collection of binary bits where MSB bits are found to be more informative. To introduce the variations in the global colour information on the palm image, the four MSB bits can be changed by keeping intact the first four bits (LSB bits). The Gaussian filter can be used to maintain the noise and sharpness values. Also, the module containing sharpness and dynamic range helps to assess amount of skin cracks and presence of shadows generated by the folds of the palm, respectively. As, sharpness represents the contrast between the edges, refers to the amount of skin cracks. Also, dynamic range is the contrast ratio between darkest and brightest colour tone where darkest colour tone refers to shadow. Histogram information containing bins concentration can also ensure the presence of shadow. Table 2 shows the ISO standardised values considered in our proposed methodology.

3.3. Extracting features from video data

Feature extraction is an essential part of pattern classification and prediction. It facilitates to extract the features by identifying the most relevant and distinctive points to be encoded in a feature set. To extract the relevant features of blood flow in palm tissue from palm video, a number of frames are extracted. The variation of blood flow occurs by applying pressure using the sphygmomanometer cuff near to wrist. To comply with the theoretical conjuncture behind the nature of blood distribution on palm image after application of pressure, the temporal correlation measurements using Pearson correlation are performed as shown in Algorithm 3. The theoretical conjuncture infers the histogram correlation of the first frame with all the remaining frames. The curve obtained from these temporal responses gives an idea of dominant blood distribution changes in palm tissues. As part of feature extraction, three vital feature points are extracted at the initial step, such as the beginning point where the pressure is applied or the point where fading

starts (P_1), the point of maximum fading (P_2), and the point of colour restoration (P_3). The polynomial curve fitting function [30] is applied to extracted correlation values, and it is then approximated to a hypothetical curve. The peaks - P_1 and P_3 represent a maximum change of direction and P_2 represents minimum change of direction, obtained from approximated correlation values' gradients [31]. These feature points are then used to determine the time and slope of de-oxygenation and re-oxygenation. Eqs. (1) and (2) denote the time formulations, whereas Eqs. (3) and (4) denote the slopes. In addition, it includes de-oxygenation time (DT), re-oxygenation time (RT), de-oxygenation slope (DS), re-oxygenation slope (RS), age, gender (Male is represented by 0 and Female by 1), correlation values of first ten frames, correlation values of five forward and five backward frames from three detected points (P_1 , P_2 and P_3), median red channel values of first ten frames each, median red channel values of each of the five forward and five backward frames distant from three detected points (P_1 , P_2 and P_3), and blurriness value which is calculated using the variance of Laplacian and the proposed edge detection operator. The edge detection proposed by us [32] is constructed from the combination of Canny and Sobel edge detection operators and texton gradient. Further, texton gradient is obtained from the synthesis of texton image on the use of filter bank with k-means clustering and pair disk mask. Fig. 4 shows the steps required for feature extraction, and Fig. 5 shows the histogram correlation graph along with approximated curve.

$$DT = \frac{\text{frame number of } p_2 - \text{frame number of } p_1}{\text{Video data frame rate}} \quad (1)$$

$$RT = \frac{\text{frame number of } p_3 - \text{frame number of } p_2}{\text{Video data frame rate}} \quad (2)$$

$$DS = \frac{\text{correlation value of } p_2 - \text{correlation value of } p_1}{\text{frame number of } p_2 - \text{frame number of } p_1} \quad (3)$$

$$RS = \frac{\text{correlation value of } p_3 - \text{correlation value of } p_2}{\text{frame number of } p_3 - \text{frame number of } p_2} \quad (4)$$

3.4. Prediction models

The proposed method is used to predict the haemoglobin levels of individuals on the use of regression models, multi-layer perceptron (MLP) network and fusion rule. The regression models are used to determine the correlation between the dependent variable (outcome) and independent variables (features). The linear regression models find the best fit lines or planes to minimise the errors, whereas non-linear regression models find the best-fit curves or hyper-planes for minimising

Algorithm 3 Temporal Correlation Measurement

Input : ROI Frames
Output : Correlation values

```

1: For each video  $crop\_frames = [Cf_1, Cf_2, \dots, Cf_N]$ 
2:  $corr\_values = []$ 
3:  $im1 = im[0]$ 
   // Read 1st cropped frame  $Cf_1$  as reference frame
4:  $H^R = \text{Extract histogram of } im1$ 
5: for  $<i = 1 : N>$  do
6:    $im2 = im[i]$ 
   // Read  $i^{th}$  frame as a candidate input frame
7:    $H_i^C = \text{Extract histogram of } im2$ 
8:    $d(H^R, H_i^C) = \frac{\sum_{l=0}^{255} (H^R(l) - \overline{H^R})(H_i^C(l) - \overline{H_i^C})}{\sqrt{\sum_{l=0}^{255} (H^R(l) - \overline{H^R})^2 \sum_{l=0}^{255} (H_i^C(l) - \overline{H_i^C})^2}}$ , where  $\overline{H_k} = \frac{1}{N} \sum_{l=0}^{255} H_k(l)$  and  $N = \text{Total number of histogram bins}$ .
9:    $corr\_values[] \leftarrow d(H^R, H_i^C)$ 
10: end for
11: return  $corr\_values$ 

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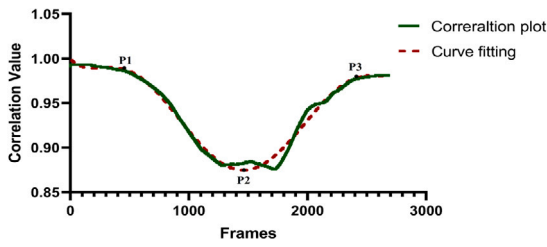


Fig. 5. Histogram correlation graph.

the errors. The experiment is performed with 12 different regression models, viz. multiple linear regression (MLR), polynomial regression (PLR), support vector regression (SVR), gradient boosting regression (GBR), Bayesian ridge regression (BRR), elastic net regression (ER), ridge regression (RR), least absolute shrinkage and selection operator regression (LAR), stochastic gradient descent regression (SGDR), K-nearest neighbors Regression (kNNR), decision tree regression (DTR) and random forest regression (RFTR). For these regression models, the clinical haemoglobin level is considered as the dependent variable, and 87 features that have been extracted, considered as independent variables.

To evaluate the performance of the proposed method, initially, the regression models are used to obtain the haemoglobin levels of individuals determined from the video of palm pallor and later on, the MLP network, also called feature-based MLP, is used to consider 87 features in the input layer for training and further, prediction of haemoglobin level. MLP can ensure quick prediction, and it also ensures better accuracy even with a smaller dataset. A single hidden layer is enough for a smaller set of features; however, multiple hidden layers are found to be more effective for higher-dimensional features. To ensure quick predictions after training and achieving higher accuracy with a smaller data set, some additional properties of MLP are exploited further with a number of hidden layers and a combination of activation function and weight updation. The chosen combination of neurons in the hidden layers of the MLP network yields the optimal convergence between the actual and predicted values.

Three different hybrid prediction models, with the help of regression models, MLP network and fusion strategy, have been proposed to enhance the ability to predict the haemoglobin level in a more precise manner. Subsequent sections present the hybrid prediction models with necessary illustrations and schematic diagrams.

3.4.1. Methodology-I

This hybrid model selects the best five regression models from a set of twelve regression models and then corresponding predicted

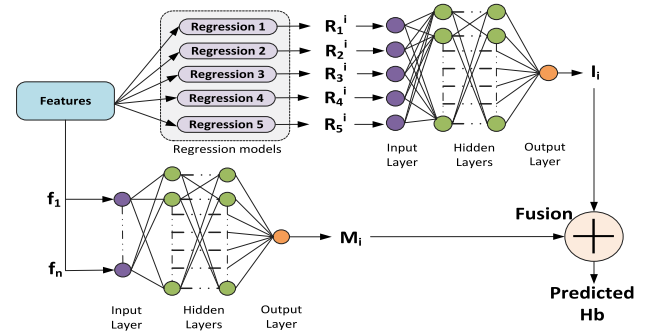


Fig. 6. Framework of Methodology-I.

values are passed through a regression-based neural network. In this network, let, R_1^i to R_5^i , for $i = 1$ to n , where n is the total number of subjects, denote the predicted haemoglobin values of all subjects in the test set, determined from the best five regression models. The input layer of the regression-based MLP, having five neurons, takes the haemoglobin levels determined from the best five regression models. There are three hidden layers in regression-based MLP. Further, the output layer consists of one neuron that generates the output I_i for each subject, and M_i represents the haemoglobin level determined from feature-based MLP for each subject. A fusion strategy called weighted sum fusion is applied to the outcomes of both the MLP networks viz. I_i and M_i are shown in Fig. 6 and combined to produce a fused and final predicted haemoglobin level (Hb_1^i). The outcome of the weighted sum rule does not contain much redundant information as well as it does not contain too little information. Hence, the weighted sum rule is found to be advantageous in score-level fusion. The fusion rule is given by

$$Hb_1^i = f(I_i, M_i) = w_1^i \times I_i + w_2^i \times M_i \quad (5)$$

$$\text{where, } w_1^i = \frac{I_i}{(I_i + M_i)} \text{ and } w_2^i = \frac{M_i}{(I_i + M_i)}.$$

3.4.2. Methodology-II

This hybrid model applies the weighted-sum fusion to the haemoglobin levels determined from the best five regression models. The weighted-sum fusion is given by

$$R_i - \text{Fusion} = w_1^i \times R_1^i + w_2^i \times R_2^i + w_3^i \times R_3^i + w_4^i \times R_4^i + w_5^i \times R_5^i \quad (6)$$

where,

$$w_1^i = \frac{R_1^i}{(R_1^i + R_2^i + R_3^i + R_4^i + R_5^i)}, w_2^i = \frac{R_2^i}{(R_1^i + R_2^i + R_3^i + R_4^i + R_5^i)}, w_3^i = \frac{R_3^i}{(R_1^i + R_2^i + R_3^i + R_4^i + R_5^i)},$$

$$w_4^i = \frac{R_4^i}{(R_1^i + R_2^i + R_3^i + R_4^i + R_5^i)}, \text{ and } w_5^i = \frac{R_5^i}{(R_1^i + R_2^i + R_3^i + R_4^i + R_5^i)}.$$

After obtaining the weighted fusion score, the regression-based MLP is applied with two inputs in the input layer, viz. weighted fusion score ($R_i - \text{Fusion}$) and the score determined from feature-based MLP (M_i). The regression-based MLP consists of three hidden layers and an output layer. The score is determined in the output layer is considered as the final predicted haemoglobin level (Hb_2^i) as shown in Fig. 7.

3.4.3. Methodology-III

This model considers the outcomes obtained from five regression-based MLP networks and the outcome obtained from feature-based MLP together. The predicted haemoglobin level determined from five regression models are again passed through another set of five regression-based MLP networks separately and obtain decouple scores at the output layers of five such networks. In the regression-based MLP networks, every network consists of the input layer with one neuron, three hidden layers and the output layer with one neuron. After obtaining five decouple haemoglobin levels at the output layers of five regression-based MLP networks, the weighted-sum fusion is applied to fuse these scores with the predicted haemoglobin level determined from

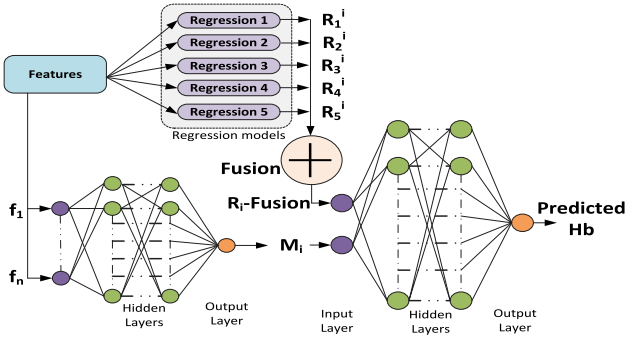


Fig. 7. Framework of Methodology-II.

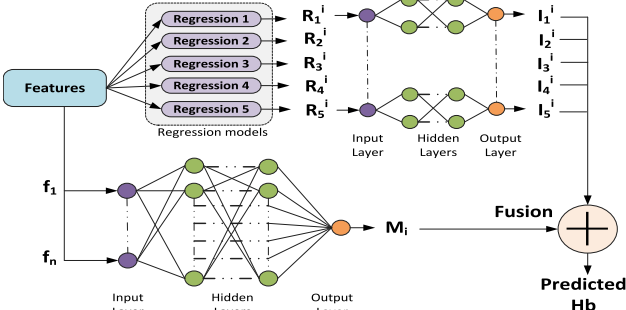


Fig. 8. Framework of Methodology-III.

the feature-based MLP network. Let, $I_1^i, I_2^i, I_3^i, I_4^i, I_5^i$ be the predicted haemoglobin levels obtained as decoupled scores from five regression-based MLP networks and M_i be the predicted score obtained from feature-based MLP network as shown in Fig. 8. The weighted-sum fusion is formulated as

$$Hb3_i = w_1^i \times I_1^i + w_2^i \times I_2^i + w_3^i \times I_3^i + w_4^i \times I_4^i + w_5^i \times I_5^i + w_6^i \times M_i \quad (7)$$

where,

$$w_1^i = \frac{I_1^i}{(I_1^i + I_2^i + I_3^i + I_4^i + I_5^i + M_i)}, w_2^i = \frac{I_2^i}{(I_1^i + I_2^i + I_3^i + I_4^i + I_5^i + M_i)}, w_3^i = \frac{I_3^i}{(I_1^i + I_2^i + I_3^i + I_4^i + I_5^i + M_i)},$$

$$w_4^i = \frac{I_4^i}{(I_1^i + I_2^i + I_3^i + I_4^i + I_5^i + M_i)}, w_5^i = \frac{I_5^i}{(I_1^i + I_2^i + I_3^i + I_4^i + I_5^i + M_i)}, \text{ and } w_6^i = \frac{M_i}{(I_1^i + I_2^i + I_3^i + I_4^i + I_5^i + M_i)}$$

4. Evaluation

This section presents the experimental results determined from the proposed method, along with the experimental setup used for evaluation and a comparison of the proposed method with state-of-the-art methods. For the experiment, the prototype of the palm device is used for capturing the video of palm pallor changes after applying and thereafter releasing the pressure. The video data are collected from a number of subjects during data acquisition. Fig. 9 shows the distribution of haemoglobin levels among different subjects. The experiment uses various performance metrics including MSE, RMSE, Bland-Altman method of 95% limits of agreement and computation of confusion matrix.

The proposed system is developed on the hypothesis that there exists an inter-relationship between the blood haemoglobin concentration of a subject with the de-oxygenation and re-oxygenation time stamps. The timestamps are evaluated from the histogram correlation graph, generated after applying constant pressure on a palm and releasing it. In an age-old practice, anaemia is detected through rough assessment of blood density on a palm by applying pressure and analysing it. The same age-old practice is exploited in the proposed system by analysing and measuring the blood haemoglobin concentration with time stamp metrics. An anaemic patient, due to poor haemoglobin concentration,

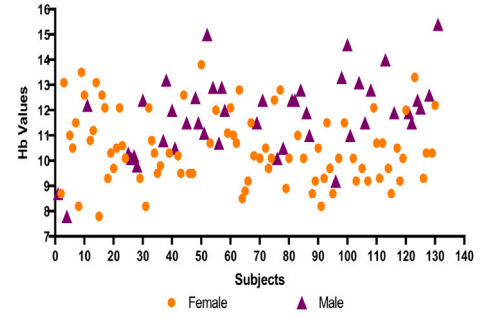


Fig. 9. Trade-off between number of subjects and corresponding clinical Hb levels.

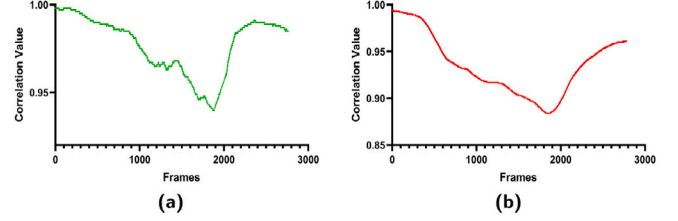


Fig. 10. Graph of (a) a non-anaemic subject. (b) an anaemic subject.

experiences paleness caused by de-oxygenation while applying pressure on the palm pallor [33]. In contrast, on releasing pressure, palm pallor takes a larger time to restore the colour caused by re-oxygenation when the poor haemoglobin concentration is evident on the same palm pallor. However, in the case of non-anaemic subjects having substantial haemoglobin concentration, the time for re-oxygenation is found to be less as compared to anaemic subjects. It has been observed that the variation in the de-oxygenation and re-oxygenation times are considered as distinct features for segregating the anaemic and non-anaemic subjects. To evaluate the various timestamp metrics relevant to the anaemia detection task, the Histogram Correlation graph can be estimated with the help of the Pearson Correlation Coefficient. Moreover, Pearson Correlation Coefficient is used to find the histogram difference of all the input frames with respect to a reference frame. Further, the Histogram Correlation graph is used to differentiate between anaemic subjects and non-anaemic subjects in terms of timestamp metrics. Fig. 10 shows a couple of graphs determined from non-anaemic and anaemic subjects, respectively.

4.1. Experimental setup

To evaluate the haemoglobin level accurately, the experiments are performed with 12 different regression models, feature-based multi-layer perceptron networks and three hybrid prediction models. To present the prediction model in robust way, multi-layer perceptron neural network is further exploited and combined with a set of distinctive features and regression models separately.

The feature-based MLP uses a maximum of three hidden layers with a sigmoid activation function and mean squared error as a function. Variations are by changing the number of hidden layers in the feature-based MLP. The input features are normalised between zero and one as the sigmoid activation function is used in layers of feature-based MLP. While a single hidden layer with 150 neurons is used in feature-based MLP, the learning rate of 0.01 is fixed with 10000 iterations. When two hidden layers are used with 150 neurons in the first layer and 80 neurons in the second layer, the learning rate of 0.09 is fixed with 10000 iterations. This arrangement of hidden layers with various other parameters produces optimal results. To improve the performance of the feature-based MLP further, three hidden layers are introduced with

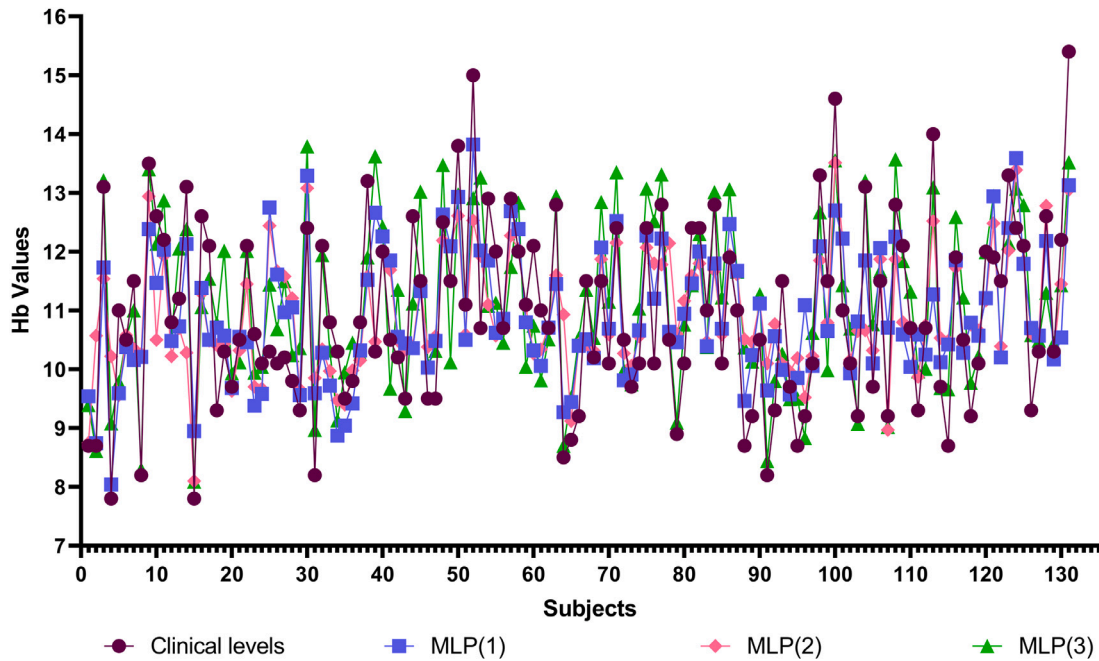


Fig. 11. Clinical levels vs predicted levels determined from feature-based MLP networks.

150 neurons in the first layer, 80 neurons in the second layer and 30 neurons in the third layer. The learning rate of 0.9 is fixed with 20000 iterations. This hierarchical structure with three hidden layers gives better predictive capability by learning the representation of features and mapping them from the lower level of representation to the higher level of representations. The number of hidden layers and the neurons in the hidden layers are chosen in such a way that the MLP network does not biased in memorising the training samples while ensuring its generalisation property. Too many connections between the hidden layers can affect the memorisation problem.

Unlike the feature-based MLP that uses three hidden layers with a sigmoid activation function, the regression-based MLP uses the ReLU activation function with mean squared error function for weight updating while learning the feature representations. In addition, the gradient descent optimisation function is used with a learning rate of 0.1 and 250 iterations while three hidden layers are used with 50 neurons in the first hidden layer, 25 neurons in the second hidden layer and 50 neurons in the third hidden layer, ensure the optimal outcome. This arrangement of regression-based MLP further ensures the best possible mapping of predicted values generated by regression models to a more robust representation using MLP towards determining the haemoglobin level.

The total number of 131 samples in the dataset is divided into 131 different test sets individually. While carrying out the evaluation, Test Set-1 considers one samples for testing and 130 samples for training. Having these train-test splits, all samples are tested, and haemoglobin labels are being predicted through an Leave-one-out cross-validation.

4.2. Experimental results

The experiments are performed with the proposed system, and experimental results are determined in accordance with various performance metrics. The proposed system estimates haemoglobin levels with the use of regression models, multi-layer perceptron networks and fusion-based predictive methodologies. The average MSE and average RMSE values from 12 regression models, three multi-layer perceptron networks and three fusion-based hybrid methodologies, are determined as shown in Table 3. These errors are determined while trying to

achieve the clinical haemoglobin levels by various methods. The regression models can predict the haemoglobin levels close to the clinical haemoglobin levels; however, due to less amount of samples, the optimal hyperplane does not accurately minimise the errors. Although, it has been observed that the non-linear regression models outperform linear ones. This observation further helps to choose the best five regression models that produce optimal MSE and RMSE values. To improve the prediction performance, 87 features are considered as input to the three variants of multi-layer perceptron networks where many hidden layers vary with neurons. Table 3 and Fig. 11 show that all three feature-based MLP variants exhibit promising results. However, the MLP variant with three hidden layers outperforms the other two variants of feature-based MLP and 12 regression models while MSE and RMSE values are computed. In addition, three variants of regression-based MLP networks are also presented to compute the average MSE and RMSE values while using the predicted haemoglobin levels determined from the best five regression models in the MLP network. However, these regression-based MLP variants depict comparable performance or less with other non-linear regression models and feature-based MLP networks. Further, to gain maximum benefits from both feature-based MLP and regression-based MLP networks and combine them to improve the prediction results, three hybrid prediction models as Methodology-I, Methodology-II and Methodology-III are proposed. The average MSE and RMSE values determined from Methodology-I outperforms not only the other two hybrid prediction models, viz. Methodology-II and Methodology-III, but also other prediction models as well, and are depicted in Table 3.

The predicted haemoglobin levels determined from three feature-based MLP networks with the corresponding clinical levels are shown in Fig. 11. The variant of feature-based MLP with three hidden layers achieves optimal MSE and RMSE values of 0.91 and 0.76, respectively, compared to other prediction models. The best five regression models include random forest regression, least absolute shrinkage and selection operator (Lasso) regression, stochastic gradient descent regression, gradient boosting regression and elastic net regression produce average RMSE values of 0.77, 0.82, 0.83, 0.84, and 0.91, respectively. The predicted haemoglobin levels determined from the proposed three hybrid prediction models as Methodologies I-III, along with intermediary results, are shown in Fig. 12. Among three proposed methodologies,

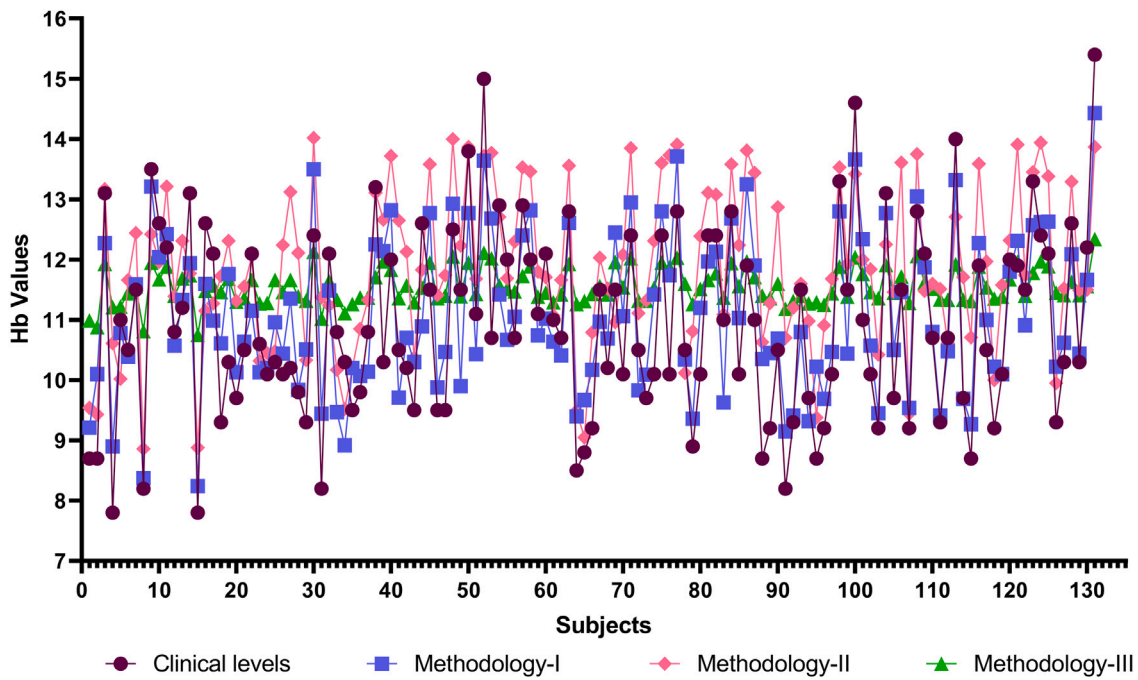


Fig. 12. Clinical levels vs predicted levels determined from hybrid prediction models.

Table 3
MSE and RMSE of different prediction models.

Prediction models	Average MSE	Average RMSE
MLR	3.87	1.51
PLR	2.95	1.08
SVR	1.33	0.96
GBR	1.03	0.84
BRR	1.22	0.92
ER	1.21	0.91
RR	1.24	0.92
LAR	0.98	0.82
SGDR	1.02	0.83
kNNR	1.41	0.98
DTR	1.82	1.15
RFTR	0.89	0.77
MLP(1)	1.16	0.89
MLP(2)	1.28	0.91
MLP(3)	0.91	0.76
Methodology-I	0.701	0.698
Methodology-II	1.04	0.86
Methodology-III	1.08	0.87

Methodology-I outperforms others, with average MSE and RMSE values of 0.701 and 0.698, respectively. Hence, Methodology-I is considered as the final prediction model for predicting the haemoglobin level of an individual more accurately.

4.2.1. Bland-Altman plot

The *Bland-Altman analysis* shows the goodness-of-fit plot between the predicted and the clinical haemoglobin levels. The bias, upper level of agreements, and lower level of agreements are plotted in Fig. 13(a). It depicts the distribution of levels within 95% LOA (limits of agreement) with a standard deviation of ± 1.96 from mean bias. The proposed hybrid prediction model as Methodology-I has a minimal bias of -0.133 g/dL and has 95% limits of agreement of $[-1.75, 1.48]$. Also, Fig. 13(b) shows the structured comparison between predicted Hb levels determined from Methodology-I and the haemoglobin levels determined from clinical measurements. As the *Bland-Altman analysis* is used for systematic comparisons of different methods in terms of gold standard, the 95% limits of the agreement shows the acceptable

range of errors determined from such different methods, provided repeatability of measurement, has been introduced. Repeatability of measurement of a number of subjects can be introduced in the proposed anaemia detection system in order to validate the performance, the limits of agreement adopted. Further, to determine the mean errors between the system generated Hb levels and clinical measurement, the *Bland-Altman analysis* ensures the usefulness of the proposed system on multiple measurements over time with acceptable error rates within the range of 95% limits of agreement. However, the proposed system restricts itself from measuring the haemoglobin levels multiple times for the same subject.

4.2.2. Risk analysis

The risk of the proposed method is assessed by presenting the trade-off between haemoglobin levels (predicted and clinical) and corresponding anaemic individuals in the dataset. The notion of risk analysis graph depicts the percentage of anaemic subjects who are being correctly classified as anaemic and, on the other hand, the remaining percentage of anaemic subjects who are being misclassified as non-anaemic. The objective is to reduce the percentage of misclassification for anaemic as well as for non-anaemic individuals out of the total sample space. According to WHO scale, in pregnant female whose $Hb < 11$ g/dL, non-pregnant female whose $Hb < 12$ g/dL and in males whose $Hb < 13$ g/dL are considered as anaemic. In the dataset, 93 subjects are found to be anaemic, among which predicted value of seven subjects is showing as non-anaemic and others as anaemic, as shown in Fig. 14. The sensitivity and specificity of the proposed system are found to be 0.93 and 0.61, respectively. The study shows that the proposed anaemia detection system tends to correctly classify the anaemic subjects as anaemic in more numbers and non-anaemic subjects as non-anaemic in fewer numbers. The classification accuracy is found to be 85.49%. If the proposed system considers the haemoglobin level less than 11.5 g/dL as anaemic for all types of subjects over different ranges as in [34], then the sensitivity, specificity and accuracy of the proposed system are determined to be 0.94, 0.81, and 88.55%, respectively. Table 4 exhibits the confusion matrix considering the WHO scale for classification of anaemic and non-anaemic. However,

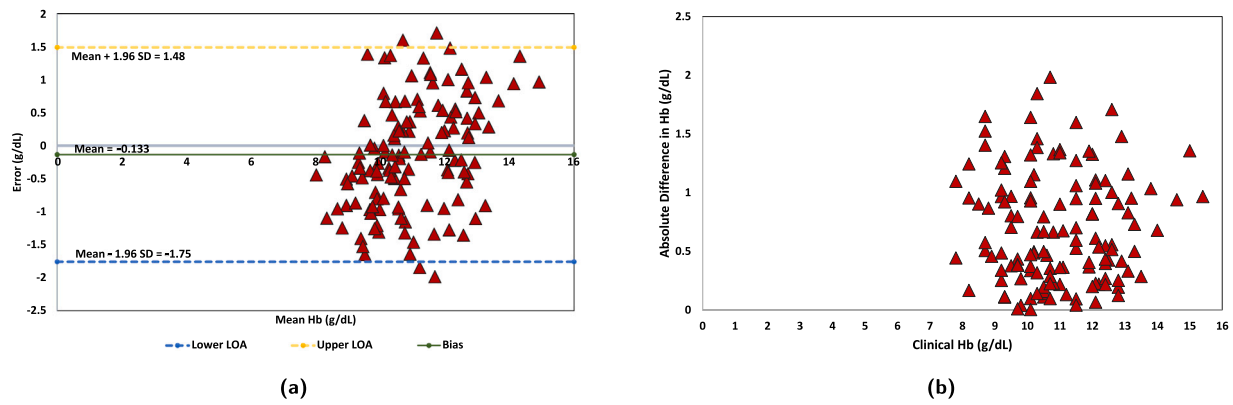


Fig. 13. (a) Bland-Altman plot with 95% limits of agreement from Methodology-I. (b) Structured comparison between predicted Hb levels from Methodology-I and the clinical level.

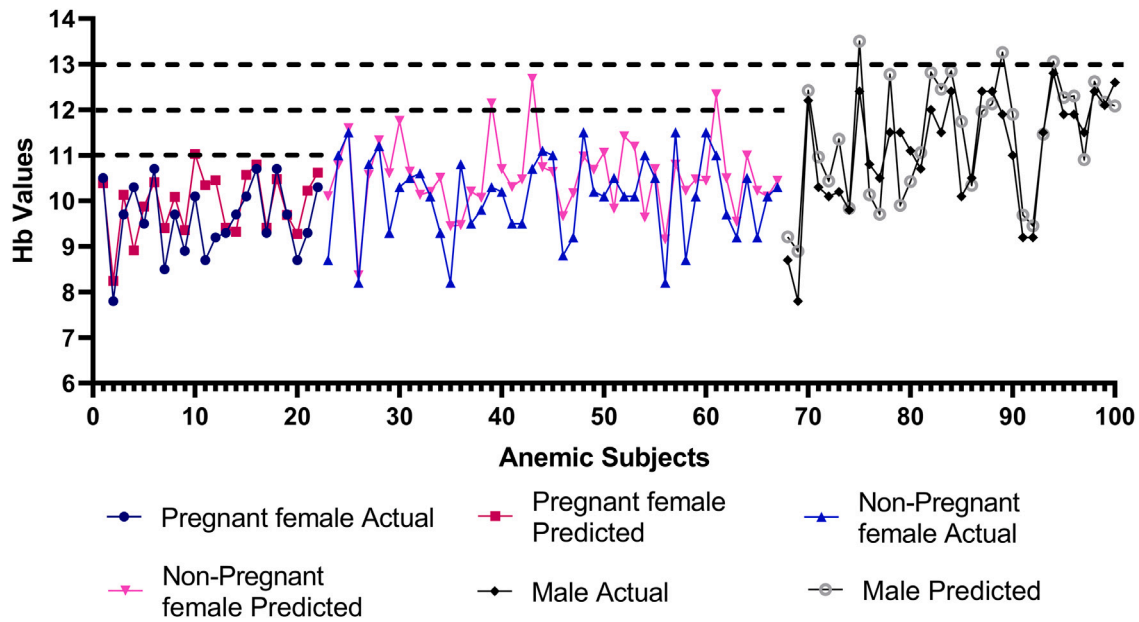


Fig. 14. Risk analysis graph.

Table 4

Confusion Matrix considering the WHO scale for classification.

Predicted condition	Actual condition		Sensitivity (TP/TP+FN)	Specificity (TN/TN+FP)
	Anaemic	Non-anaemic		
Anaemic	93 (TP)	7 (FN)	0.93	0.61
Non-anaemic	12 (FP)	19 (TN)		

Table 5

Confusion Matrix considering a threshold of 11.5 g/dL for classification.

Predicted condition	Actual condition		Sensitivity (TP/TP+FN)	Specificity (TN/TN+FP)
	Anaemic	Non-anaemic		
Anaemic	73 (TP)	5 (FN)	0.94	0.81
Non-anaemic	10 (FP)	43 (TN)		

the same is represented in Table 5 considering a threshold of 11.5 g/dL as the anaemic/non-anaemic classification.

4.3. Comparison with different state-of-the-arts

This section presents a comparison of the proposed system with some state-of-the-art methods. Although, the proposed system uses

different physiological evidence for estimating the haemoglobin levels of individuals with existing methods. But, the comparison with the proposed system exhibits a credible study about the performance of the anaemia detection systems. Moreover, the comparison reveals how the various performance metrics are used to evaluate the systems more accurately while considering different experimental setups and the nature of environments. Only a few research works are found in the

Table 6
Comparison with different existing methods.

Papers	Evidence	Sample size	Age	Hb range	Gender ratio (M:F)	LOA	Bias	RMSE	MSE	Sensitivity	Specificity	Accuracy	Correlation	Mean error
HemaApp [12]	Non-invasive (Finger tip)	31	6 to 77 (mean=31)	8.3–15.8	15:16	3.6	–	–	–	85.70%	76.50%	–	0.69	1.26
Masimo Pronto [10]	Non-invasive (Finger tip)	18	–	7.5–14	–	2.76	0.14	–	–	69.30%	88.20%	–	–	1.28
Masimo co-oximetry [9]	Non-invasive (Finger tip)	20	20 to 27 (mean=24)	7.5–13.8	11:09	3.7	–0.15	0.94	–	–	–	–	–	–
M. K. Hasan [35]	Non-invasive (Finger tip)	75	20 to 56	–	20:55	–	–	–	–	94%	96%	92%	0.93	–
G. M. Ahsan [36]	Non-invasive (Finger tip)	84	–	–	–	–	–	–	–	–	–	48.8%	0.56	–
Mannino [13]	Non-invasive (Nail bed)	337 (train=237 & test=100)	1 to 60	5.9–16.8	170:167	2.4	0.2	–	–	92%	76%	–	–	0.97
S. Acharya [37]	Non-invasive (Nail)	1466	–	–	–	1.96	0.019	1.353	–	–	–	–	–	–
R. Dsh [38]	Non-invasive (Nail)	1005	–	–	–	–	–	1.47	–	–	–	–	0.79	–
A. Tamir [39]	Non-invasive (Eye Conjunctiva)	19	–	6.9–16	7:12	3.5	–	–	–	–	–	78.9%	–	–
S. Collings [40]	Non-invasive (Eye conjunctiva)	47	–	–	–	3.5	–	–	–	74%	71%	–	–	–
Muthalagu[41]	Non-invasive (Eye conjunctiva)	127	–	–	–	–	–	–	–	77%	96%	–	–	–
Sagnik Ghosal[34]	Non-invasive (Eye conjunctiva)	65	16 to 80 (mean=42)	7.5–14.5	36:24	0.9	0.15	–	–	89%	–	–	–	0.32
G. Dimauro[42]	Non-invasive (Eye conjunctiva)	65	20 to 80	–	47:18	–	–	–	–	–	–	–	0.74	–
Bikash Santra [11]	Non-invasive (Palm Pallor)	20	23 to 50	9.6–13.7	15:05	–	–	1.144	–	–	–	–	0.563	–
Proposed	Non-invasive (Palm Pallor)	131	18 to 79 (mean=38.09)	7.8–15.4	45:86	1.75	–0.133	0.698	0.701	93% 93.6%	61.29% 81.13%	85.49% 88.55%	0.85	0.70

literature [11,21,24] that have been considered the palm pallor for estimation of blood haemoglobin level, however with not so satisfactory outcome. Hence, the proposed system target to devise a radically new approach, i.e., non-invasive estimation of blood haemoglobin level by analysing palm pallor with high accuracy. As the proposed system uses palm pallor evidence for estimating the haemoglobin level done through a comprehensive acquisition process, certain preprocessing steps and robust prediction models, the same evidence is observed in work [11]. However, the work [11] uses different sets of features determined from palm pallor's red and green channel and a simple form of prediction model without using essential image preprocessing operations. Due to this structure, the system often fluctuates the performance with fewer data samples. In comparison to this system, the proposed system exhibits robust performance with an average RMSE value of 0.698. The comparison considers the evaluation of the anaemia detection systems with a range of performance metrics described in Table 6. The proposed system uses all such metrics to understand the performance in a more comprehensive way. In contrast, the existing ones do not use all the performance metrics for evaluation. Hence, it is quite difficult to have a clear idea about the robustness of the systems. For example, the HemaApp work [12] did not mention the metrics, such as bias, RMSE, MSE and accuracy as part of the evaluation. However, the work showed a few performance metrics, such as LOA, sensitivity, specificity, correlation and mean error. These measurements might be found to be useful while estimating haemoglobin levels. However, without important measurements like bias, it would be difficult to understand the system's repeatability while collecting the evidences from the same set of individuals over the time. These sorts of issues might not prevail with the proposed system as there is a scope for introducing different forms of constraints to improve the system further and make it a more advanced one. Similar to HemaApp work [12], the performances of other existing methods can be observed in [9–11,13,34–42], without estimating the similar set of metrics as that of the proposed one as well as among themselves. The proposed system has the ability to detect the anaemic cases more accurately and non-anaemic cases moderately despite having the presence of a few false-positive cases. The existing systems do not show this consistent strength to accurately classify the anaemic and non-anaemic cases,

with a tendency to address the anaemia detection problem in a more exhaustive way. The Table 6 contains two sensitivity, specificity and accuracy values based on WHO scale and threshold of 11.5g/dL for classification of anaemic and non-anaemic. The comparison in Table 6 clearly shows that the proposed system outperforms other state-of-the-art systems while considering the performance in terms of RMSE, MSE and mean error. Moreover, limits of agreement (LOA), bias, sensitivity, specificity, accuracy and correlation are found to be competitive and even comparable to the superior systems.

Cost Comparison

To understand the cost-effectiveness of the proposed system, it is hereby compared with the cost of existing devices available in market. The proposed device is more cost-effective as compared to the other available devices. As, the proposed device requires manual blood pressure measuring kit (costing 17 USD approximately) along with one video capturing box (costing 18 USD approximately) which is having velcro strips on the base for fixation of palm while applying constant pressure and LED light strips to provide constant illumination of the entire palm throughout the duration of the experiment. However, camera sensor of personal smartphone is used for capturing the video data of palm. A comparative study on cost comparison of the available anaemia detection technologies is shown in Table 7 wherein, the approximate cost of the proposed device as 35 USD is shown. And, if include the price of a smartphone, that might roughly cost around 140 USD. However, for a large-scale production, the market price of the product may even be lower.

5. Conclusion and future scope

This paper has presented a smartphone-based non-invasive anaemia detection system using palm evidence that provides a point-of-care test facility to individuals for monitoring their haemoglobin levels. The proposed system estimates the blood haemoglobin level by analysing the palm pallor. The image preprocessing steps help to enhance the estimation performance while considering correlation-based features. It uses the smartphone and hardware device to capture the video of palm pallor changes. The proposed system ensures the accurate estimation of haemoglobin levels with the help of video processing techniques

Table 7

Cost comparison of different anaemia detection technologies.

Technologies	Cost (approximately)
HemoCue 201 [43]	698 USD
Diaspect [44]	225 USD
HemoControl [45]	425 USD
URIT-12 [46]	159 USD
Pronto-7 [47]	629 USD
Proposed	35 USD (only device) 140 USD (including smartphone)

and hybrid fusion-based prediction models. Following the WHO scale for classification of anaemic conditions, the proposed work reports a sensitivity rate of 93% with an accuracy of 85.49% while achieving the average MSE, average RMSE and bias of 0.701, 0.698 and -0.133, respectively. In addition, 95% of limits of agreement (LOA) is estimated to be the range of [-1.75, 1.48]. Although the proposed system shows higher sensitivity with an acceptable range of errors and classification accuracy, however, it gives out to test the repeatability of the samples which are collected from the same individuals over a long period of time. Hence, addressing this issue along with the use of a large number of samples in the experiment for estimating the haemoglobin levels is a future concern. Moreover, demography information will be considered to be derived from the experiments while estimating the blood haemoglobin levels from the individuals covering diversified regions. Apart from this consideration, other health-related issues will also be taken care of to infer the decisions about certain types of symptoms related to anaemic conditions to be detected in the subjects.

CRedit authorship contribution statement

Abhishek Kesarwani: Conceptualization, Methodology, Writing – original draft, Visualization. **Sunanda Das:** Software, Methodology, Validation. **Mamata Dalui:** Supervision, Methodology, Draft correction, Project administration, Funding acquisition, Data Curation. **Dakshina Ranjan Kisku:** Supervision, Methodology, Draft correction, Formal analysis. **Bibhash Sen:** Draft correction, Writing – Review and Editing. **Suchismita Roy:** Draft correction, Writing – Review and Editing. **Anupam Basu:** Suggestions, Draft correction, Writing – Review and Editing.

Declaration of competing interest

One or more of the authors of this paper have disclosed potential or pertinent conflicts of interest, which may include receipt of payment, either direct or indirect, institutional support, or association with an entity in the biomedical field which may be perceived to have potential conflict of interest with this work. For full disclosure statements refer to <https://doi.org/10.1016/j.bspc.2022.104045>. Mamata Dalui reports financial support was provided by MeitY, GOI. Mamata Dalui has patent pending to NIT Durgapur.

Data availability

The data that has been used is confidential.

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